Tutorial 4: Systems of Linear Equations

13 October 2025

1. For each of the following systems of linear equations: (i) express the linear system as a matrix product $\boldsymbol{A}\vec{x} = \vec{b}$; (ii) perform Gaussian elimination to reduce the augmented matrix $(\boldsymbol{A} \mid \boldsymbol{b})$ into row-echelon / reduced row-echelon form, and (iii) determine the solution(s), if any, to the linear system.

   (a) $\begin{cases} x + 2y - 4z = -4 \\ 2x + 5y - 9z = -10 \\ 3x - 2y + 3z = 11 \end{cases}$

   (b) $\begin{cases} x_1 + 2x_2 - 3x_3 + 4x_4 = 2 \\ 2x_1 + 5x_2 - 2x_3 + x_4 = 1 \\ 5x_1 + 12x_2 - 7x_3 + 6x_4 = 3 \end{cases}$

   (c) $\begin{cases} x_1 - 3x_2 + 2x_3 - x_4 + 2x_5 = 2 \\ 3x_1 - 9x_2 + 7x_3 - x_4 + 3x_5 = 7 \\ 2x_1 - 6x_2 + 7x_3 + 4x_4 - 5x_5 = 7 \end{cases}$

2. (a) Consider the following linear system:

$$\begin{cases} x + 2y - 3z = 4 \\ 3x - y + 5z = 2 \\ 4x + y + \left(a^2 - 14\right) z = a + 2 \end{cases}.$$

   For what values of a will the system have (i) no solution? (ii) exactly one solution? (iii) infinitely many solutions?

   (b) Consider the following linear system:

$$\begin{cases} ax + y = a \\ x + y + z = 1 \\ y + az = b \end{cases}$$

   For what values of a and b will the system have (i) no solution? (ii) exactly one solution? (iii) infinitely many solutions?

3. 今有牛二、羊五，以十三豕，有一千。牛三、豕三，以九羊，足。羊六、豕八，以五牛，不足六百。牛、羊、豕各何?

   〈方程〉，《九章算》

   We have the following problem from Chapter 8 of *The Nine Chapters on the Mathematical Art*:

   > If you sell two cows and five sheep and you buy 13 pigs, you gain 1,000 coins. If you sell three cows and three pigs and buy nine sheep, you break even. If you sell six sheep and eight pigs and you buy five cows, you lose 600 coins. What is the price of a cow, a sheep, and a pig?

4. (a) Find a formula for the quadric surface $ax^2 + by^2 + cz^2 = d$ that passes through the points $(1, 1, 1)$, $(1, 3, 3)$, and $(-2, 0, 2)$.

   (b) The equation of a circle of radius r centered at the point (a, b) is given by $(x - a)^2 + (y - b)^2 = r^2$. Find the equation of the circle that passes through the points $(5, 5)$, $(4, 6)$, and $(6, 2)$.

5. Consider the standard basis vectors of $\mathbb{R}^3$, written as column matrices:

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

   (a) Let $\boldsymbol{A}$ be an arbitrary 3×3 matrix. What are $\boldsymbol{A}\vec{e}_1$, $\boldsymbol{A}\vec{e}_2$, and $\boldsymbol{A}\vec{e}_3$?

   (b) Let $\boldsymbol{B}$ be an arbitrary $n \times 3$ matrix, with columns $\vec{v}_1$,$\vec{v}_2$, and $\vec{v}_3$. What are $\boldsymbol{B}\vec{e}_1$, $\boldsymbol{B}\vec{e}_2$, and $\boldsymbol{B}\vec{e}_3$?

   (c) In $\mathbb{R}^n$, we define the ith standard basis vector as the $n \times 1$ matrix:

$$\vec{e}_i = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \begin{array}{l} \\ \\ \\ \leftarrow i\text{th component} \\ \\ \\ \end{array}$$

   If $\boldsymbol{A}$ is an $m \times n$ matrix, what is $\boldsymbol{A}\vec{e}_i$?

   (d) Find a 3×3 matrix $\boldsymbol{A}$ such that

$$\boldsymbol{A}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \boldsymbol{A}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \quad \boldsymbol{A}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 10 \\ 12 \end{pmatrix}.$$

6. A linear system of the form $\boldsymbol{A}\vec{x} = \vec{0}$ is called *homogeneous*. Explain why each of the following statements are true.

   (a) All homogeneous systems are consistent.

   (b) A homogeneous system with fewer equations than unknowns has infinitely many solutions.

   (c) If $\vec{x}_1$ and $\vec{x}_2$ are solutions of the homogeneous system $\boldsymbol{A}\vec{x} = \vec{0}$, then, $\vec{x}_1 + \vec{x}_2$ is a solution as well.

   (d) If $\vec{x}$ is a solution of the homogeneous system $\boldsymbol{A}\vec{x} = \vec{0}$, then, $k\vec{x}$ is a solution as well for any constant $k \in \mathbb{R}$.

7. Suppose that the linear system $\boldsymbol{A}\vec{x} = \vec{b}$ has a solution given by $\vec{v}$.

   (a) Suppose that $\vec{x}_h$ is a solution to the homogeneous system $\boldsymbol{A}\vec{x} = \vec{0}$. Show that $\vec{v} + \vec{x}_h$ is a solution of the system $\boldsymbol{A}\vec{x} = \vec{b}$.

   (b) If $\vec{w}$ is another solution of the system $\boldsymbol{A}\vec{x} = \vec{b}$, show that $\vec{w} - \vec{v}$ is a solution of the system $\boldsymbol{A}\vec{x} = \vec{0}$.

   (c) Suppose that $\boldsymbol{A}$ is a 2×2 matrix whose entries are not all zero. Furthermore, suppose that $\boldsymbol{A}\vec{x} = \vec{0}$ has at least one solution, $\vec{y} \neq \vec{0}$.

      i. Explain why the set of all solutions to $\boldsymbol{A}\vec{x} = \vec{0}$ forms a line through the origin in $\mathbb{R}^2$.

      ii. Describe the set of all solutions to $\boldsymbol{A}\vec{x} = \vec{b}$, where $\vec{b} \in \mathbb{R}^2$.